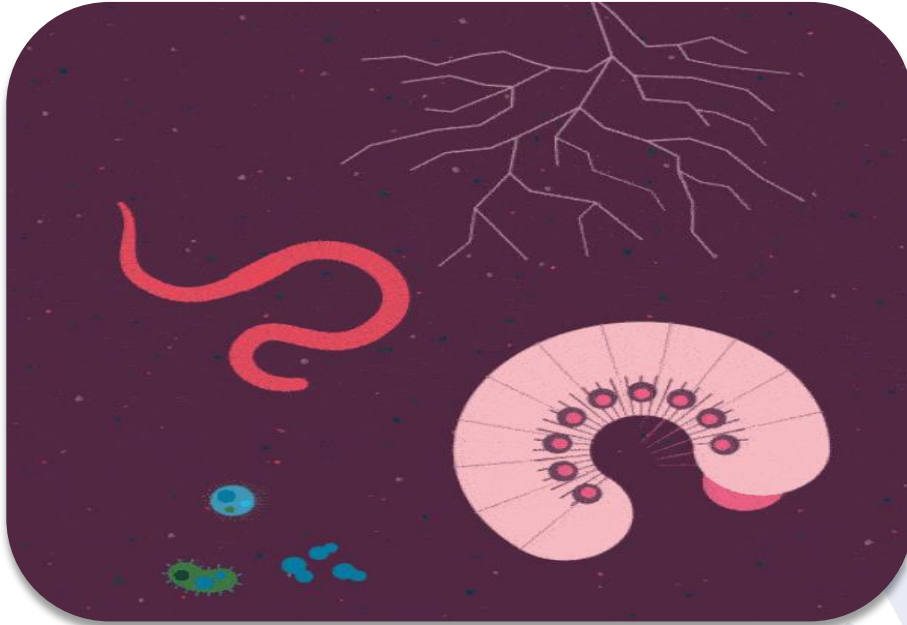


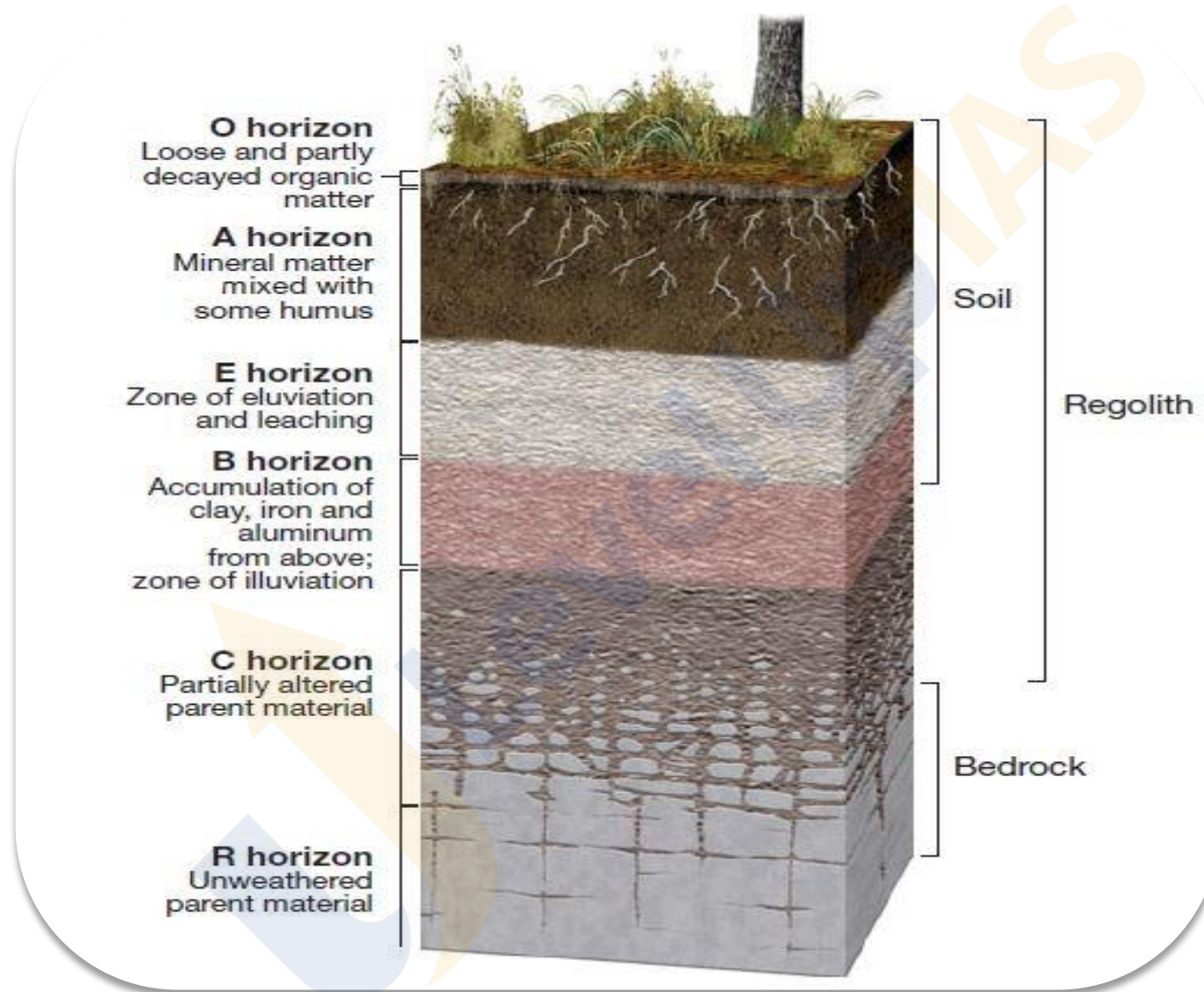
Soil

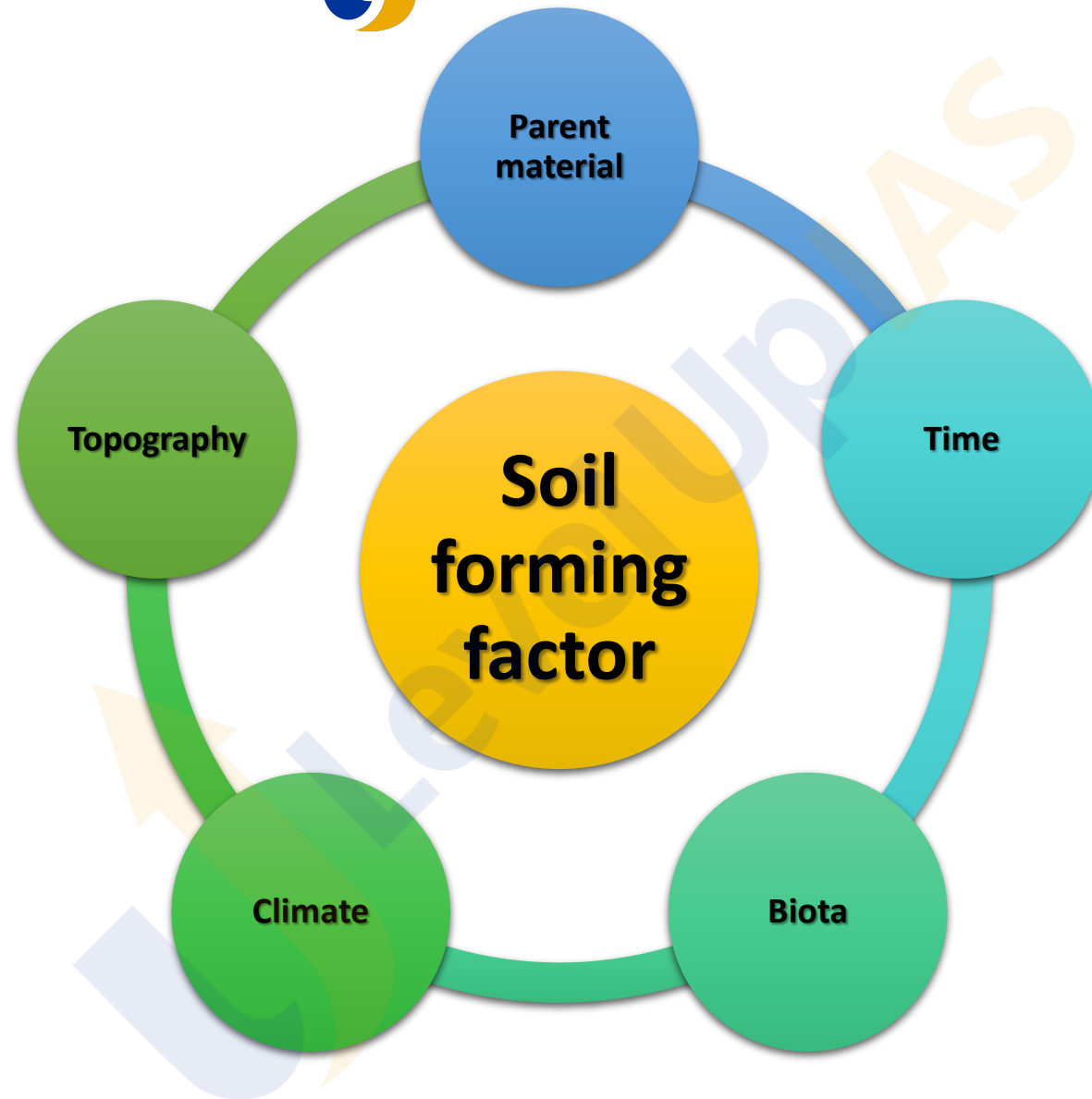
- **Weathered Parent Rock:** Weathering
- **Factors:** 5 major factors
- **Process:** Weathering, Leaching, Capillary Action.

Soil as a Ecosystem

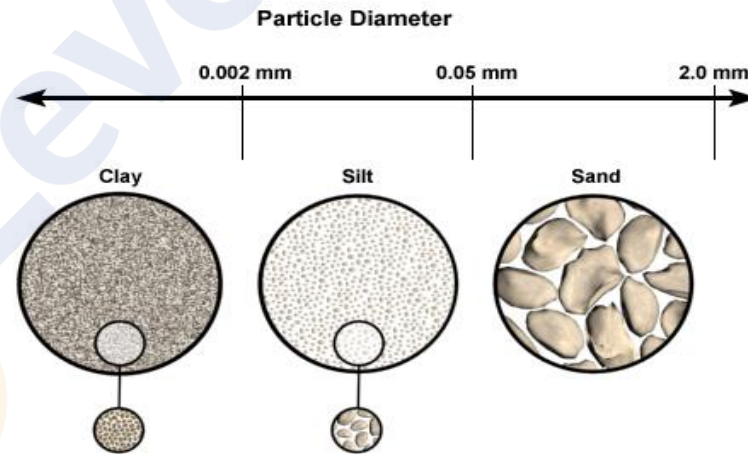


Soil Profile and Soil Horizon

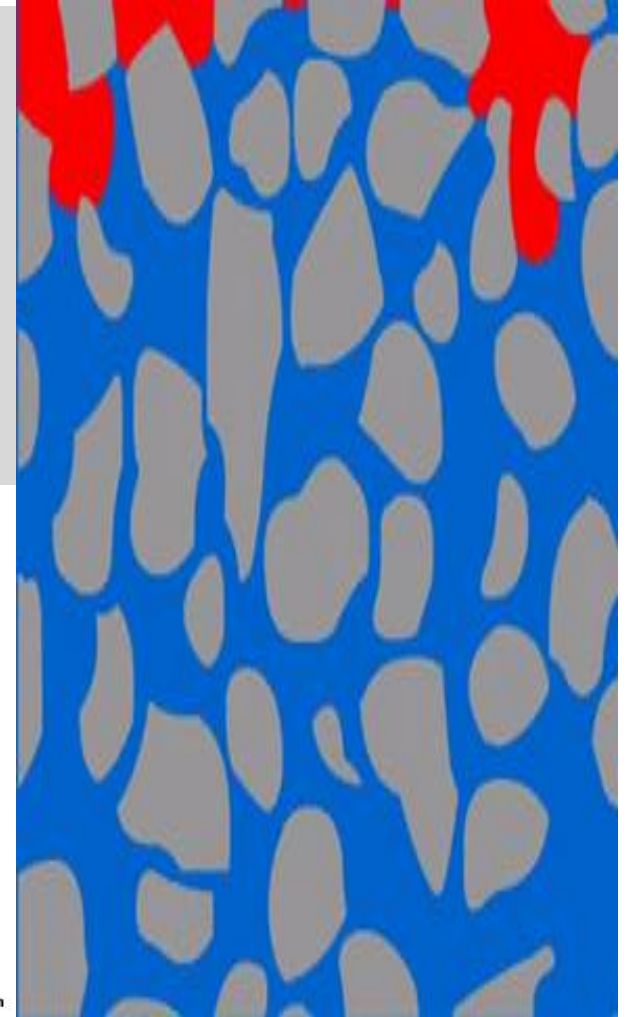


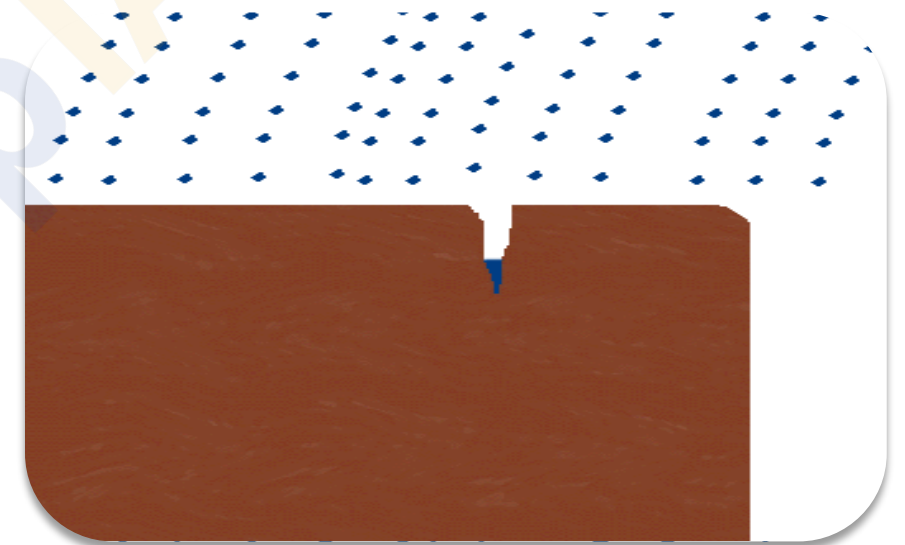
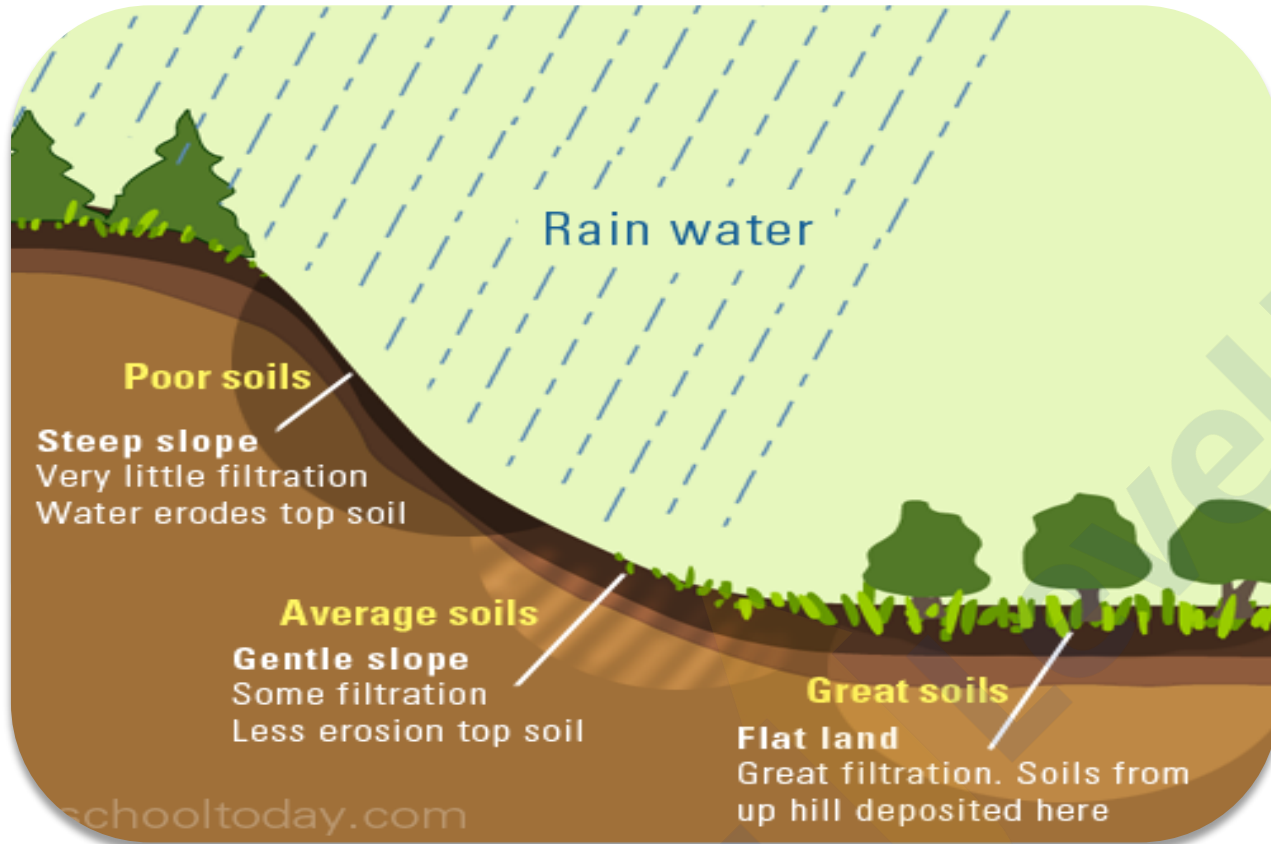


Parent material



©The COMET Program

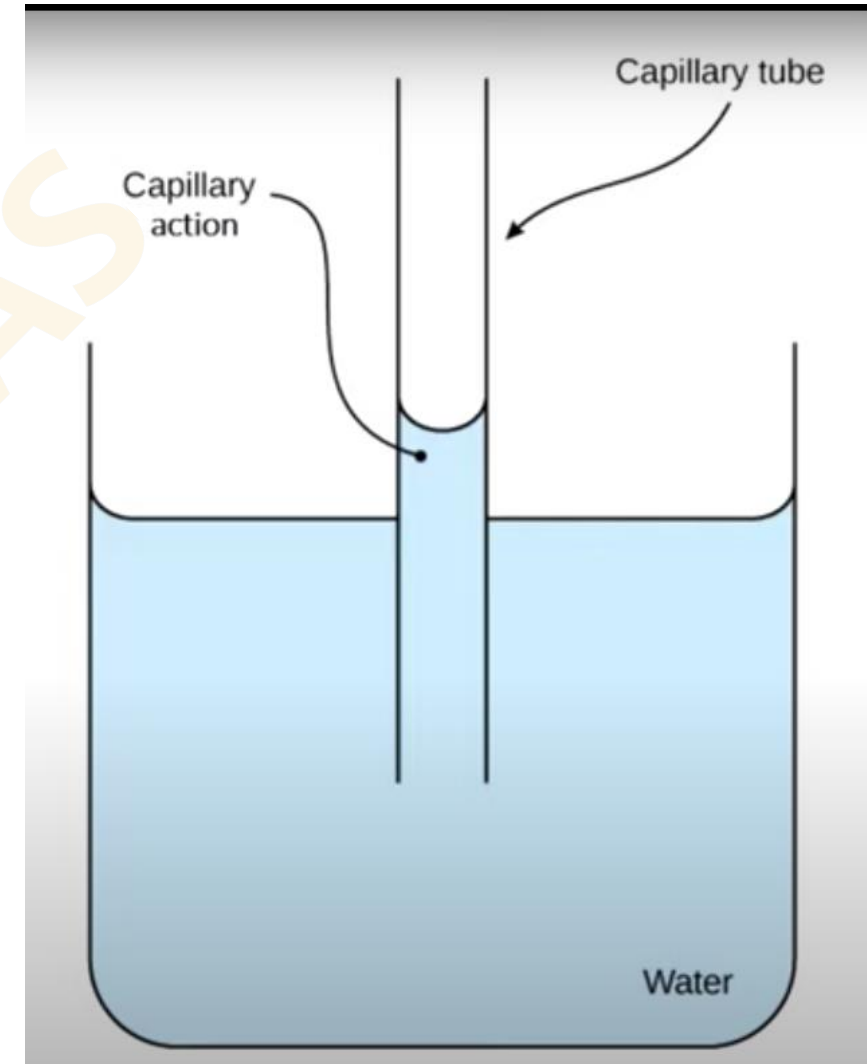
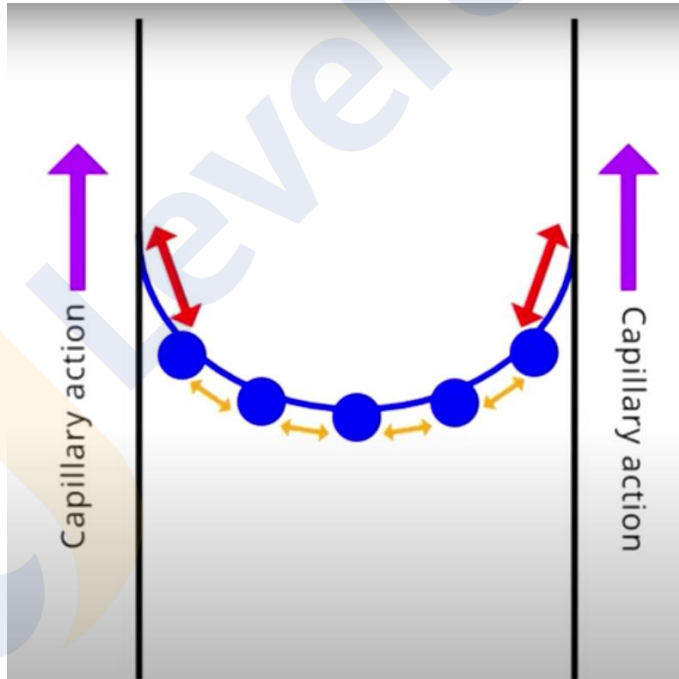




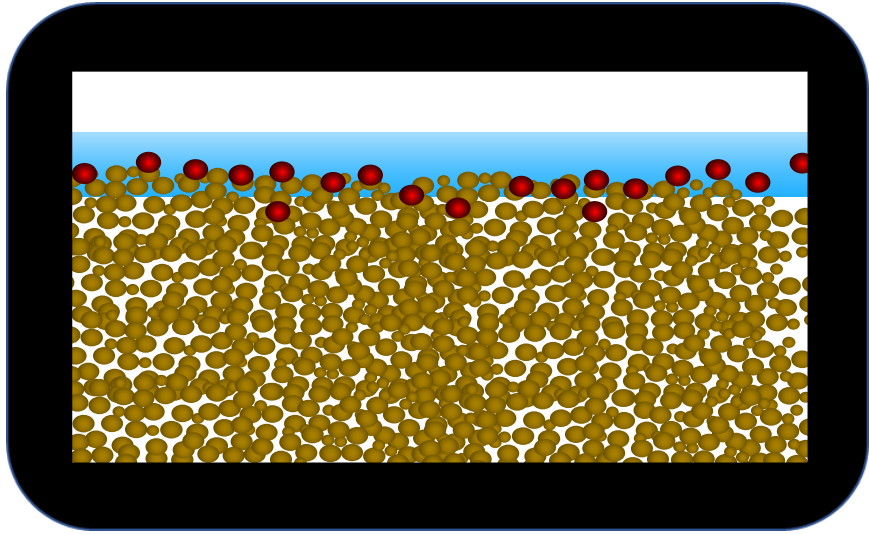
Freeze Thaw action

Capillary Action

- It is the ability of a liquid to flow through narrow spaces of a solid material without the help of external force. The flow can be against the gravity to a certain extent
- Drawing up of liquid occurs as a result of intermolecular attraction between the liquid and surrounding solid surface



Climate



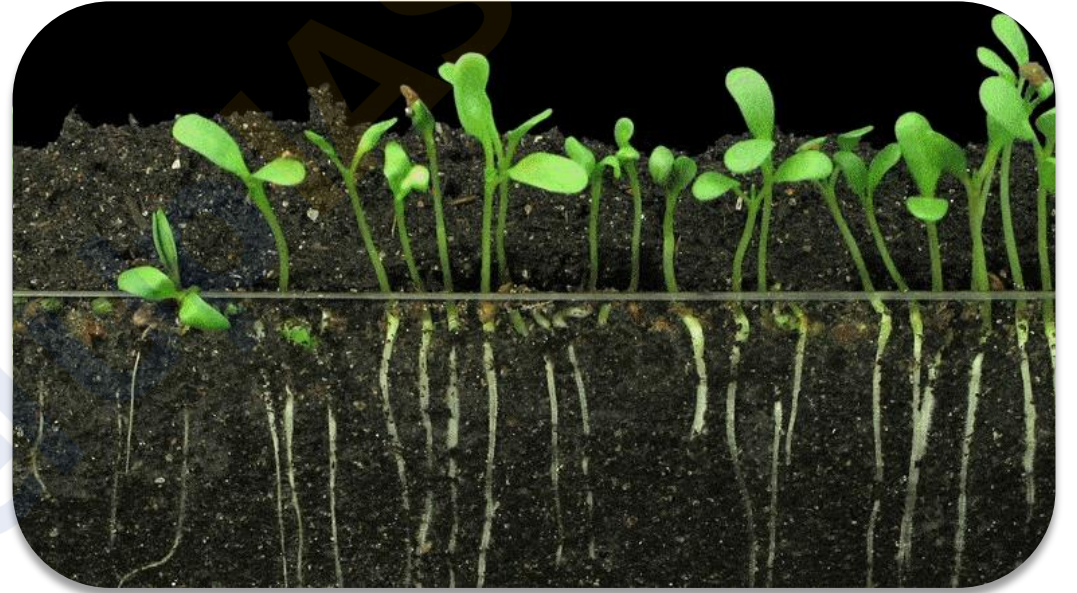
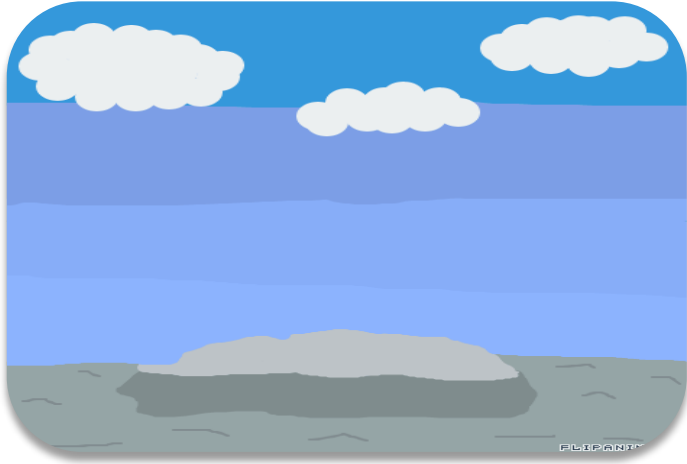
Leaching

Laterite
Leaching

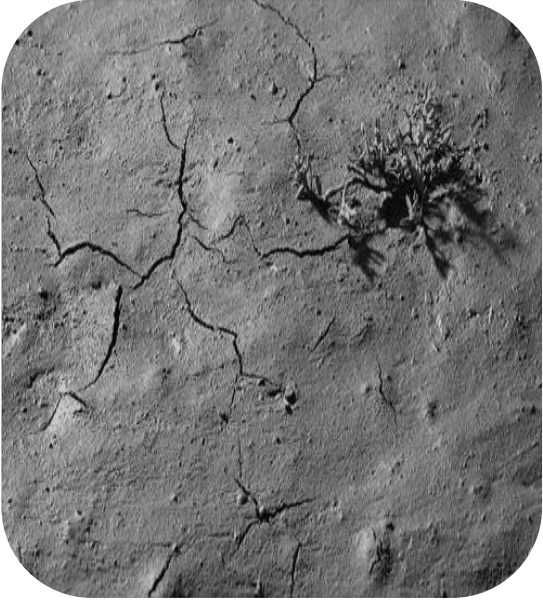


Silica leached down

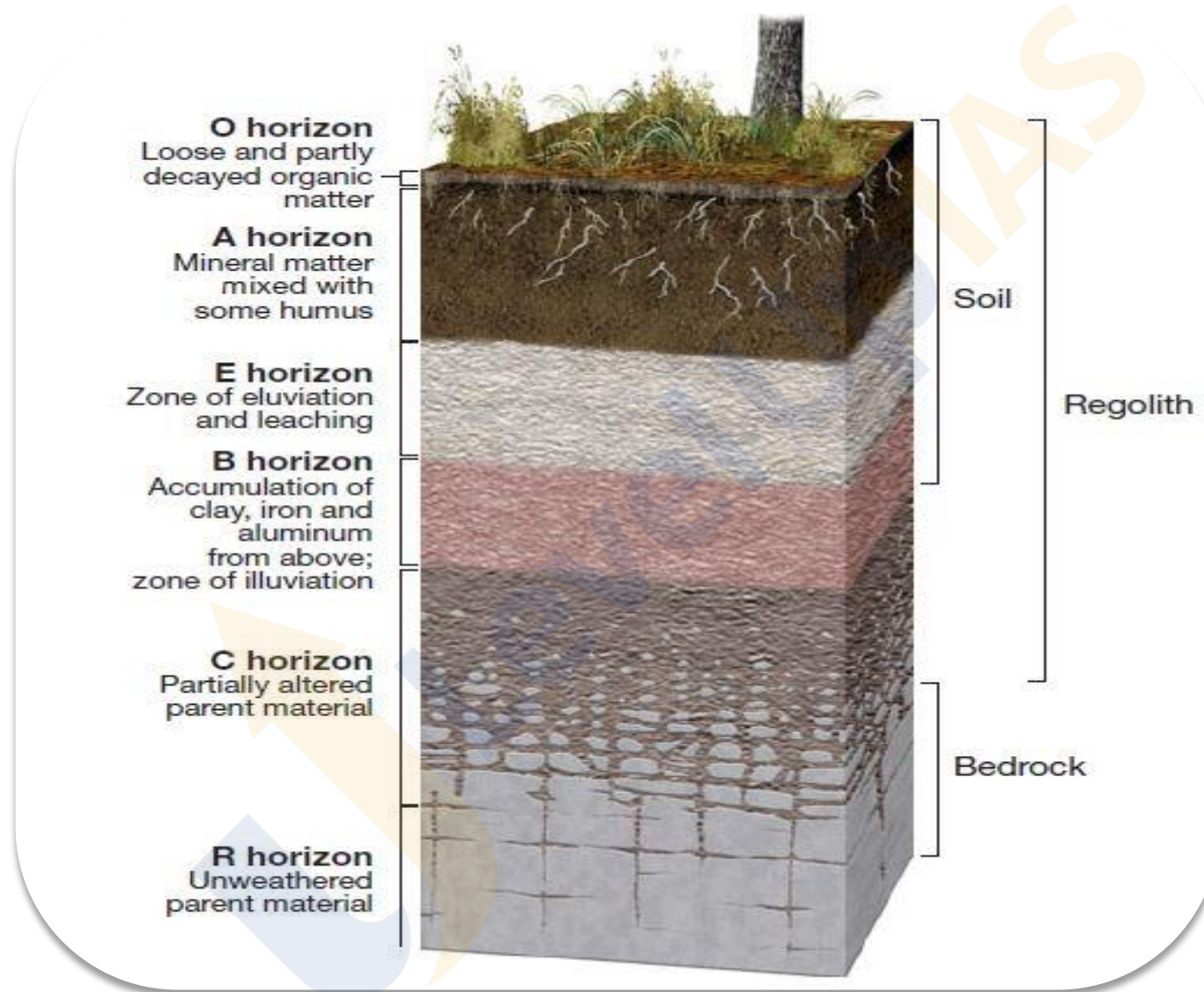
Biota



Time



Soil Profile



Indian Soils

Major Soils

Alluvial

Red

Black

Laterite

Minor soils

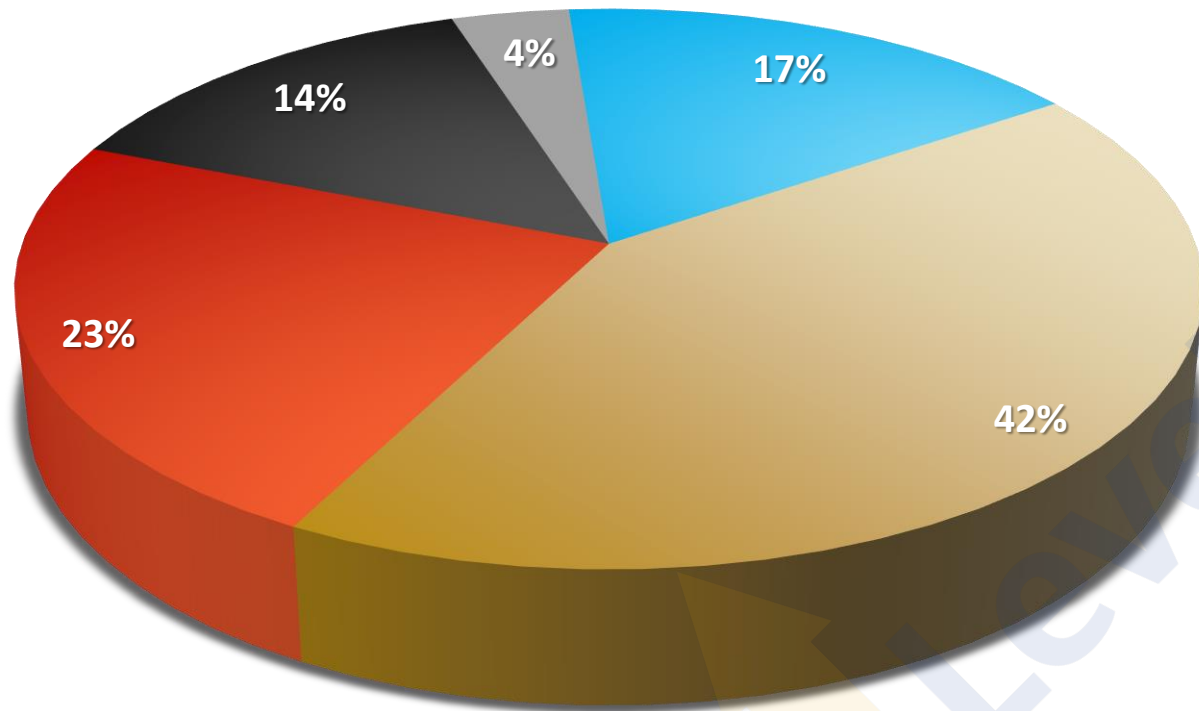
Desert

Mountain

Mangrove

Alkaline

Soils of India



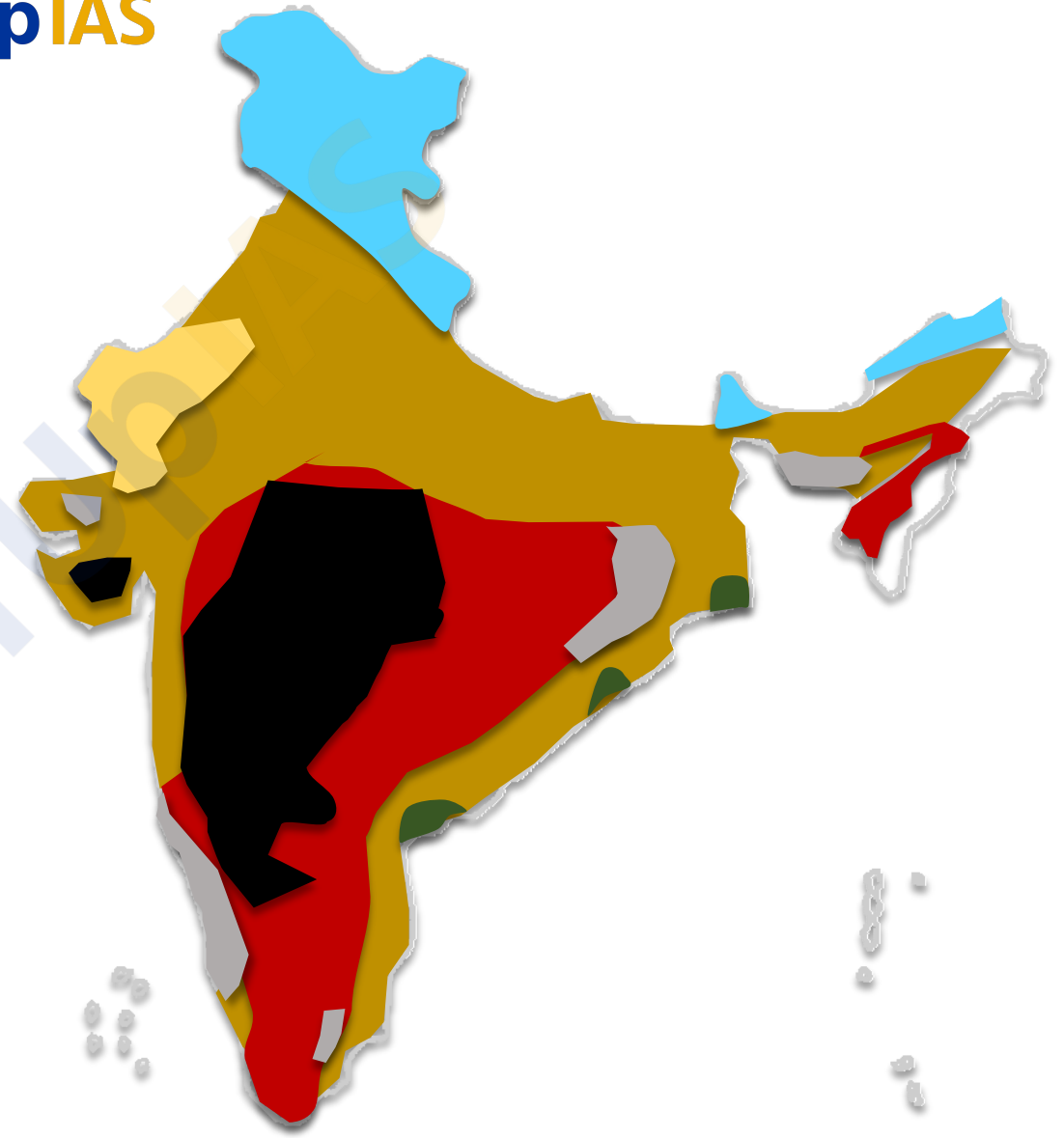
 Alluvial soil

 Red Soil

 Black Soil

 Laterite soil

 Minor soils



Alluvial Soil

Transported soil

Deposited by the river

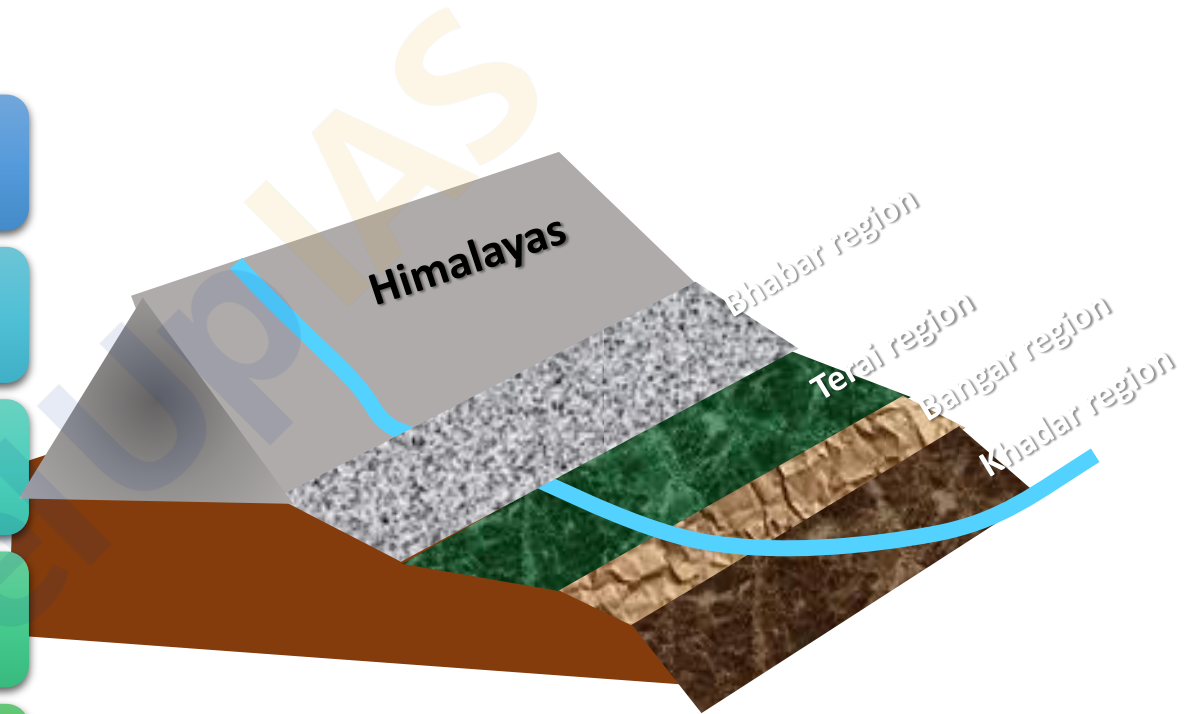
Rich in Potash

Poor in Phosphorus

Found in River basin, Valleys and deltas

Supports all types of crops.

Over cultivation, sand mining



Red Soil

Most common in Peninsular region

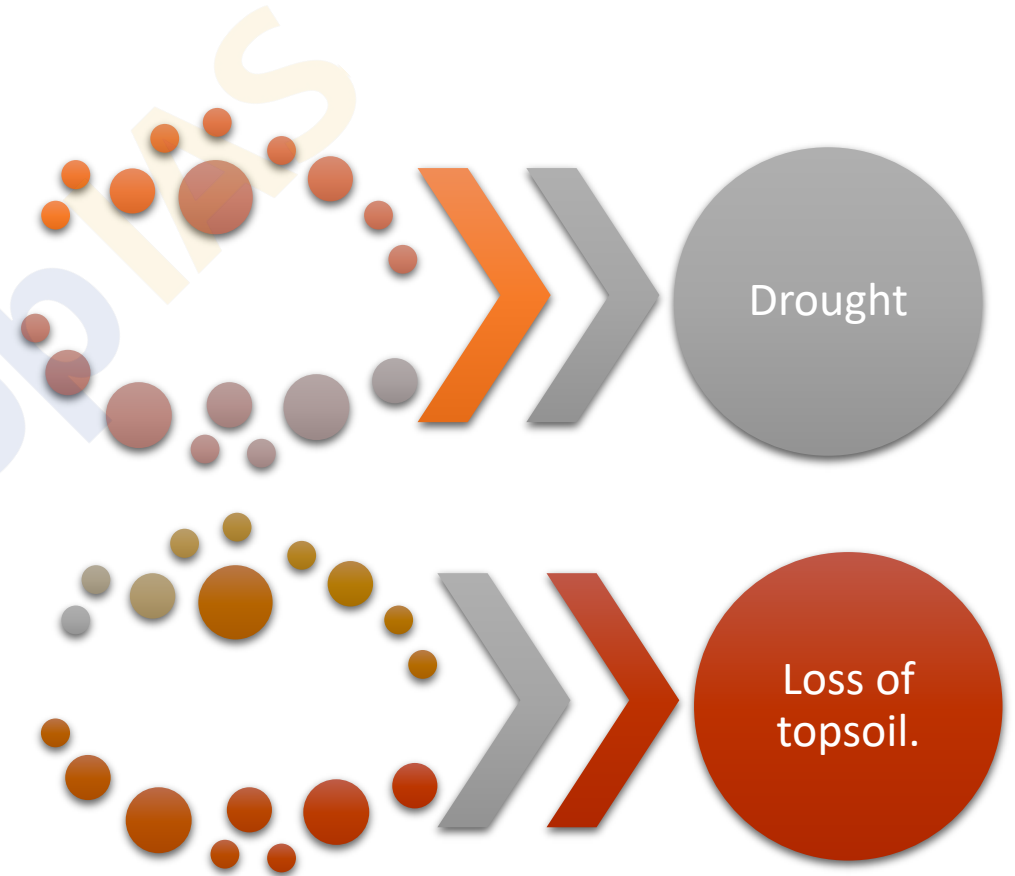
From weathering of Gneissic rock

Poor in nitrogen, phosphorus and humus

Dry soil with coarse texture (Fit for agriculture with fertilizers)

Supports dry crops – Oilseeds, Pulses Millets, Tobacco, etc.

Found in Telangana, Karnataka, Bundelkhand, T.N Plain/Plateau



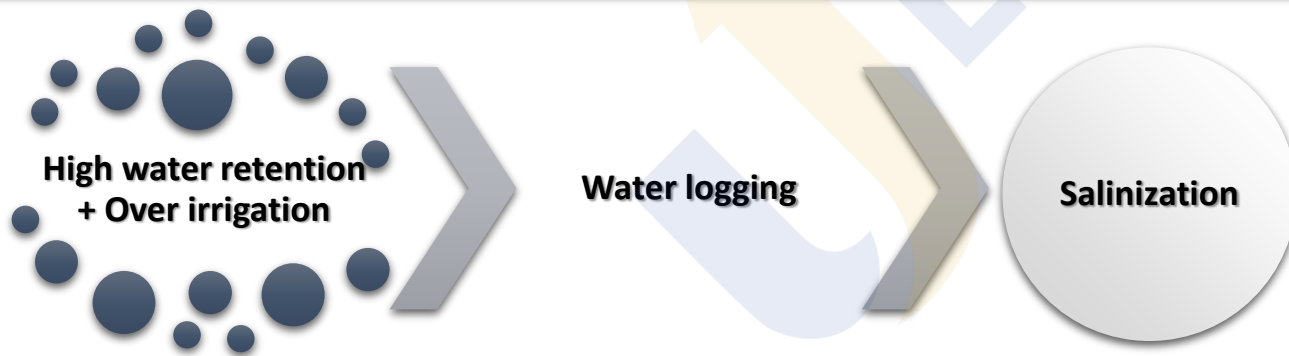
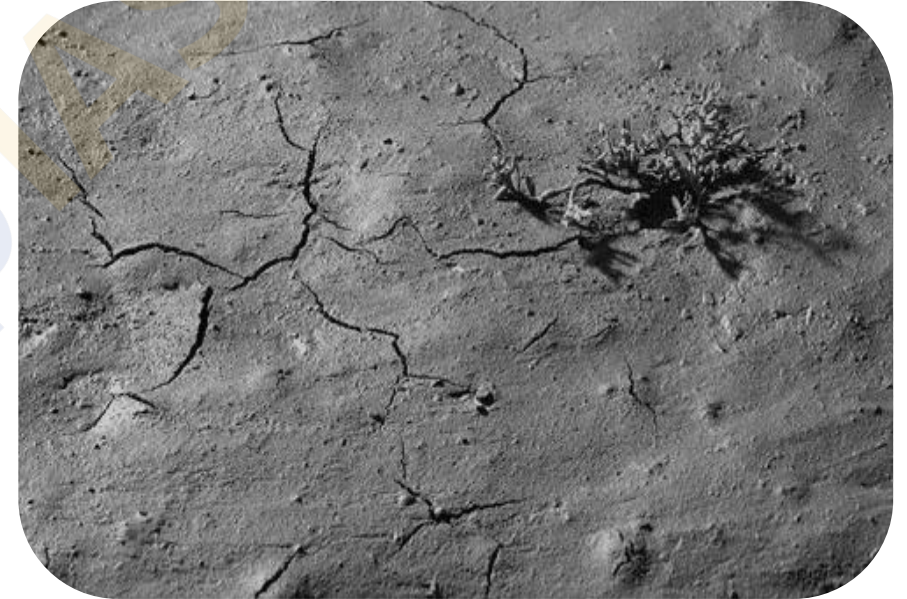
Black Soil

Soil of deccan region

From weathering of basalt rock

Found in Maharashtra, Kathiawar, South MP, Telangana, Karnataka pt.

Rich in Lime, Iron, Magnesia, Alumina



Laterite Soil

In Highland with Hot and humid climate

Through the process of laterization → leaching

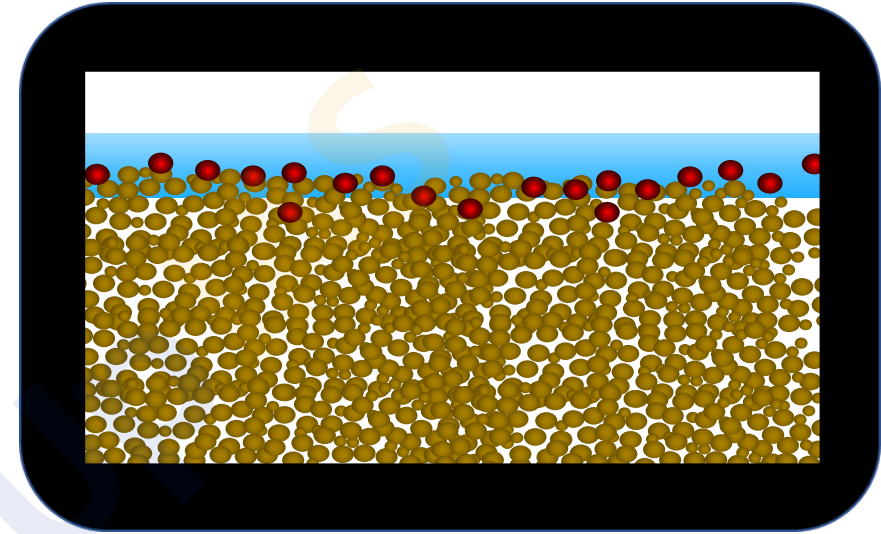
Red in colour

Fit for forestry and plantation crops.

Poor soil for agriculture.

Used mostly for brick making, Aluminum industries

Formation of IRON PAN



Forest soil

Mountain Region

Parts of the Podsol

Best for forestry

Arid Soil

Rajasthan

Poor in humus

Kankar Formation:
Problem

With irrigation these are
fit for the agriculture

Peaty soil

Along the coast

Black and sticky

Water logged

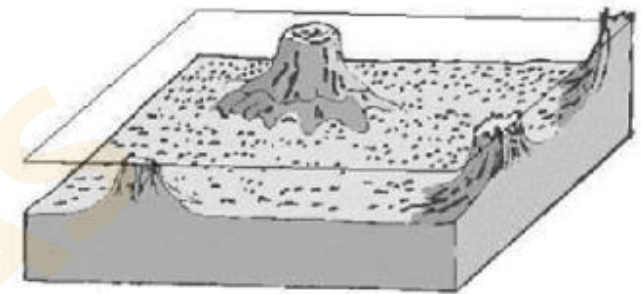
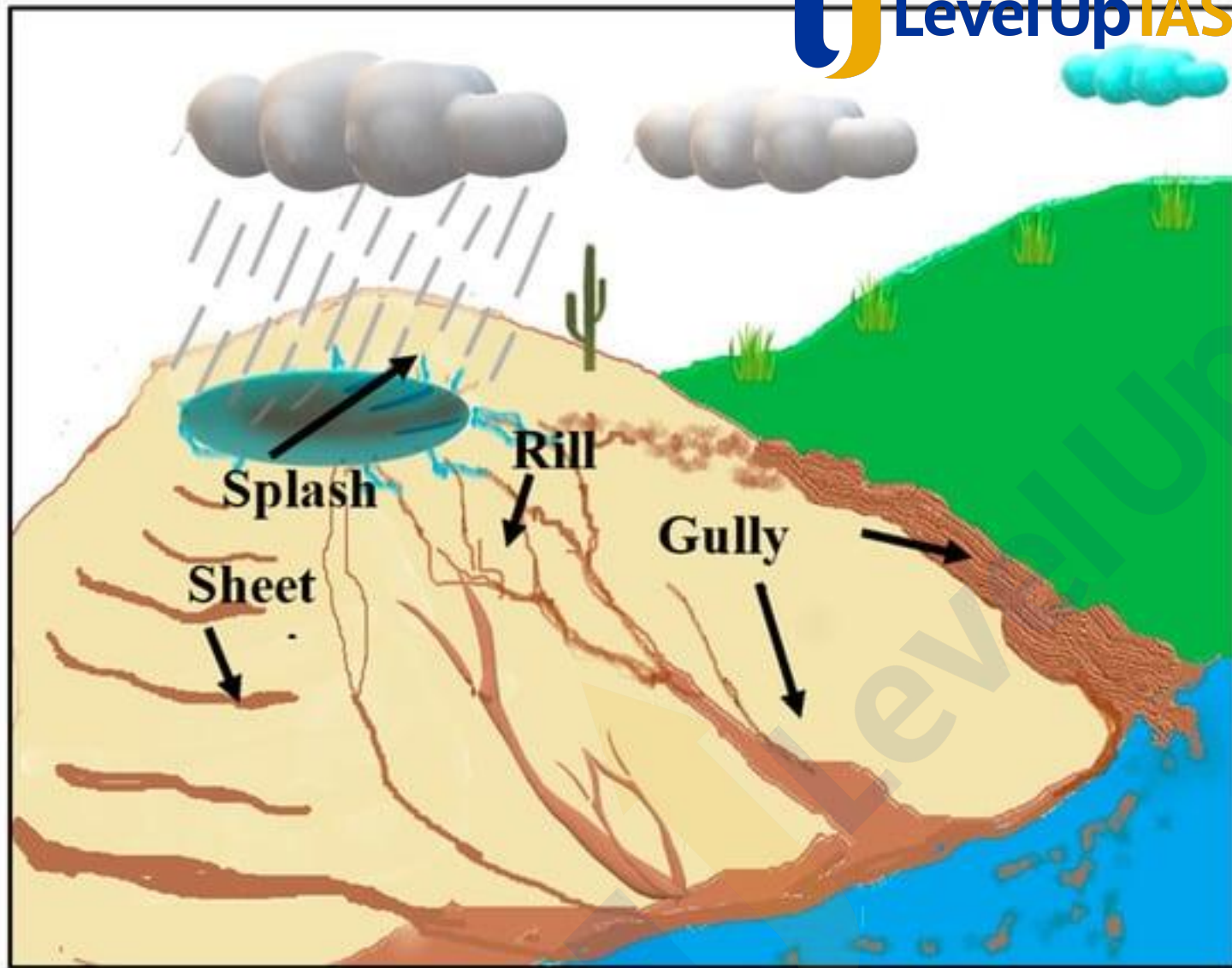
High peat content.

Saline soil

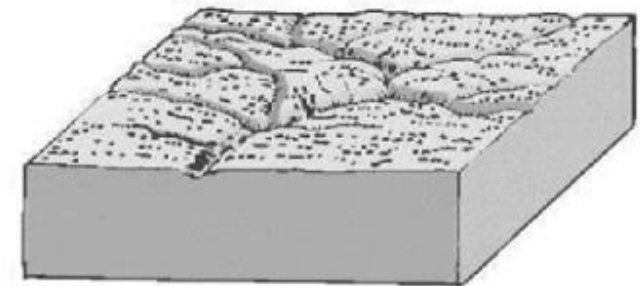
Degraded soils (Over
irrigation)

Found in poor drained
and Dry region, also in
water logged regions

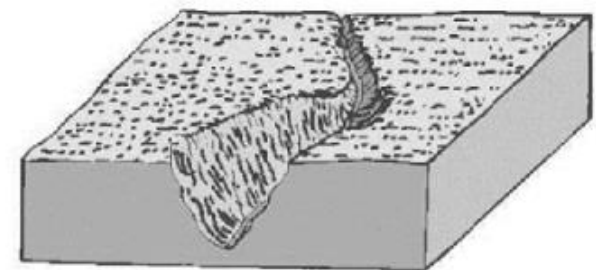
Sundarban, Gujarat



(a) Sheet erosion



(b) Rill erosion

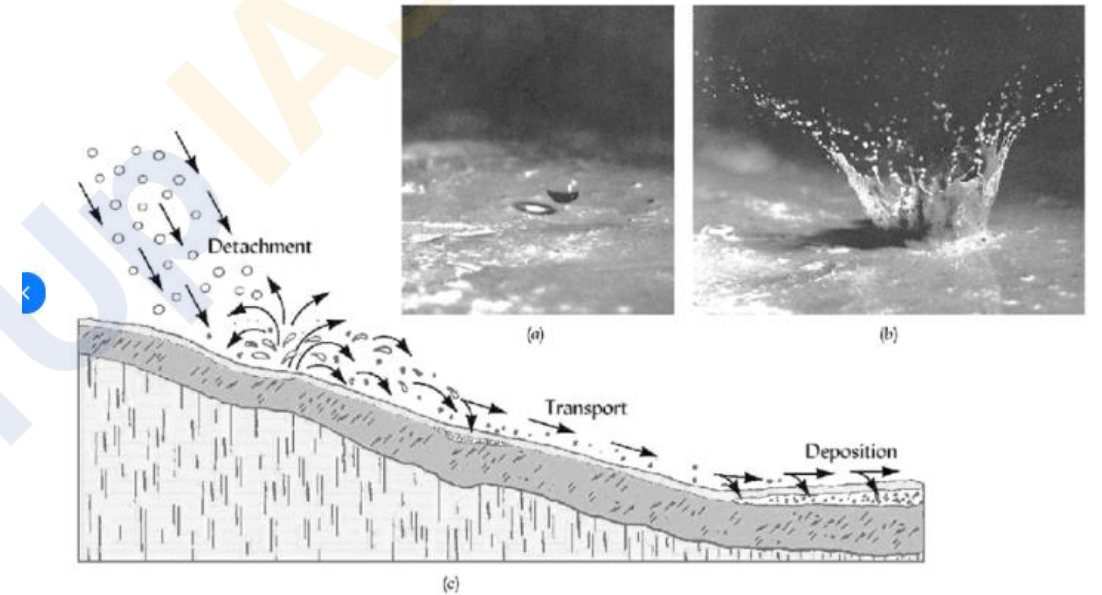


(c) Gully erosion

SOIL PROBLEMS

1. Soil Erosion:

- Removal of fertile top layer. It is associated with mining, wrong agricultural practices like fallow land (bare land without any crop cover), deforestation, over grazing.
- Fallowing of land exposes soil to hitting of raindrop leading to splash erosion. Continuous erosion by raindrop leads to erosion of entire surface in form of sheet known as sheet erosion.
- Surface water flows that cause sheet erosion rarely travel more than a few metres before turning into rills and causing Rill Erosion
- Rill eventually widen to become gully erosion



1 (a) Raindrop falling on the surface (b) Splash impact of raindrop (c) Process of water erosion (modified from Sticher 2010) 12.2.1.3. Rill erosion (channel erosion)

SOIL PROBLEMS



2. Soil Degradation: It refers to decline in productivity and quality of soil. It can be due to:

- **Soil Pollution:** Excessive use of chemical fertilizers, pesticides etc
Waste Dumping, Mining, Disposal of industrial effluents
- **Wrong Agricultural Practices** like monoculture, unscientific rotation of crops, salinisation due to cultivation of water intensive crops in dry regions, Shifting Cultivation.
- Capillary action leads to accumulation of calcium carbonate beneath (kankar), the soil will act as impermeable to water and water gets logged as happened in Indira Gandhi canal regions of Rajasthan.

3. Soil Desertification: Extreme form of soil degradation where soil is no longer fit for cultivation

- **Reasons:**

- Spread of desert like conditions along the productive soil.
- Extreme form of Land degradation: Same points as degradation but with more intensity

Solutions



- **Soil organic matter** should be maintained as it provides fertility, better water retention and holds the soil together
- **Plant trees, shrubs** to provide wind protection and also binds soil to roots.
- **Mulching:** Covering on ground of hay, straw and crops residue which prevents soil from erosion and preserves moisture in soil. It also prevents growth of weeds
- **Low Till/Minimum Tillage Methods in dry region** to conserve the soil of moisture, reduce erosion and does not disturb the soil
- **Erosion-reducing grazing practises:** Rotational grazing is a method of moving livestock from one pasture paddock to the next. Each paddock is given a rest period and is allowed to regrow naturally, reducing soil compaction and erosion.
- **Terracing esp in hilly areas** to prevent ploughing against the slope.
- **Windbreaks (also known as shelterbelts)** are rows of trees and shrubs planted along the edges of agricultural fields to provide wind protection
- **Organic farming, Precision farming** to reduce the excessive use of fertilizers, pesticide
- **Growing climate suitable crops** to prevent soil degradation

2011

23. If a tropical rain forest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because

- (a) the soil of rain forest is deficient in nutrients
- (b) propagules of the trees in a rain forest have poor viability
- (c) the rain forest species are slow-growing
- (d) exotic species invade the fertile soil of rain forest

2012

**W h i c h o n e o f t h e f o l l o w i n g i s t h e
c h a r a c t e r i s t i c c l i m a t e o f t h e
T r o p i c a l S a v a n n a h R e g i o n ?**

- (a) R a i n f a l l t h r o u g h o u t t h e y e a r
- (b) R a i n f a l l i n w i n t e r o n l y
- (c) A n e x t r e m e l y s h o r t d r y s e a s o n
- (d) A d e f i n i t e d r y a n d w e t s e a s o n

2012

Consider the following statements : If there were no phenomenon of capillarity

1. It would be difficult to use a kerosene lamp
2. One would not be able to use a straw to consume a soft drink
3. The blotting paper would fail to function
4. The big trees that we see around would not have grown on the Earth

Which of the statements given above are correct?

- (a) 1, 2 and 3 only
- (b) 1, 3 and 4 only
- (c) 2 and 4 only
- (d) 1, 2, 3 and 4

2012

C o n s i d e r t h e f o l l o w i n g a g r i c u l t u r a l p r a c t i c e s :

1. C o n t o u r b u n d i n g
2. R e l a y c r o p p i n g
3. Z e r o t i l l a g e

I n t h e c o n t e x t o f g l o b a l c l i m a t e c h a n g e , w h i c h o f t h e a b o v e h e l p s / h e l p i n c a r b o n s e q u e s t r a t i o n / s t o r a g e i n t h e s o i l ?

- (a) 1 a n d 2 o n l y
- (b) 3 o n l y
- (c) 1 , 2 a n d 3
- (d) N o n e o f t h e m

2013

Which of the following is/are unique characteristic/characteristics of equatorial forests?

1. Presence of tall, closely set trees with crowns forming a continuous canopy
2. Coexistence of a large number of species
3. Presence of numerous varieties of epiphytes

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

2013

"Climate is extreme, rainfall is scanty and the people used to be nomadic herders." The above statement best describes which of the following regions?

- (a) African Savannah
- (b) Central Asian Steppe
- (c) North American Prairie
- (d) Siberian Tundra

2013

Which of the following statements regarding laterite soils of India are correct?

1. They are generally red in colour.
2. They are rich in nitrogen and potash.
3. They are well-developed in Rajasthan and UP.
4. Tapioca and cashew nuts grow well on these soils.

Select the correct answer using the codes given below.

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 1 and 4

2014

If you travel through the Himalayas, you are likely to see which of the following plants naturally grow there?

1. Oak
2. Rhododendron
3. Sandalwood

Select the correct answer using the code given below

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

2015

"Each day is more or less the same, the morning is clear and bright with a sea breeze; as the Sun climbs high in the sky, heat mounts up, dark clouds form, then rain comes with thunder and lightning. But rain is soon over.

"Which of the following regions is described in the above passage?

- (a) Savannah
- (b) Equatorial
- (c) Monsoon
- (d) Mediterranean

2015

In a particular region in India, the local people train the roots of living trees into robust bridges across the streams. As the time passes, these bridges become stronger. These unique 'living root bridges' are found in

- (a) Meghalaya
- (b) Himachal Pradesh
- (c) Jharkhand
- (d) Tamil



2015

Consider the following States:

1. Arunachal Pradesh
2. Himachal Pradesh
3. Mizoram

In which of the above States do
'Tropical Wet Evergreen Forests'
occur?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

2015

In India, in which one of the following types of forests is teak a dominant tree species?

- (a) Tropical moist deciduous forest
- (b) Tropical rainforest
- (c) Tropical thorn scrub forest
- (d) Temperate forest with grasslands

2018

Which of the following leaf modifications occur (s) in the desert areas to inhibit water loss?

1. Hard and waxy leaves
2. Tiny leaves
3. Thorns instead of leaves

Select the correct answer using the code given below:

- (a) 2 and 3 only
- (b) 2 only
- (c) 3 only
- (d) 1, 2 and 3

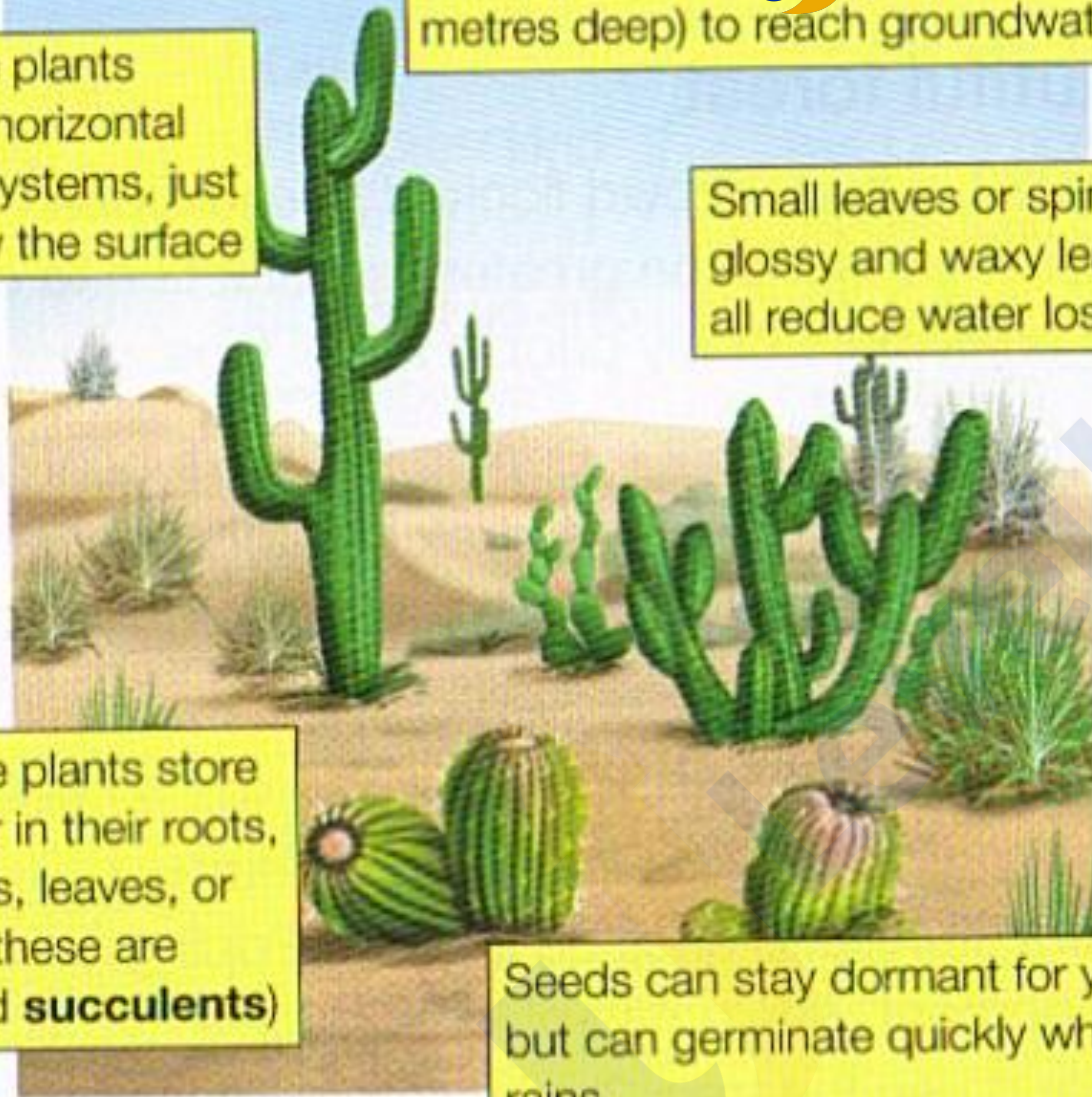
Some plants have horizontal root systems, just below the surface

Some plants have long taproots (10 metres deep) to reach groundwater

Small leaves or spines, glossy and waxy leaves all reduce water loss

Some plants store water in their roots, stems, leaves, or fruit (these are called **succulents**)

Seeds can stay dormant for years, but can germinate quickly when it rains



2018

With reference to agricultural soils, consider the following statements :

1. High content of organic matter in soil drastically reduces its water holding capacity.
2. Soil does not play any role in the sulphur cycle.
3. Irrigation over a period of time can contribute to the salinization of some agricultural lands.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

2021



The vegetation of savannah consists of grassland with scattered small trees. The forest development in such areas is generally kept in check by one or more or a combination of some conditions. Which of the following are such conditions?

1. Burrowing animals and termites.

2. Fire

3. Grazing herbivores

4. Seasonal rainfall

5. Soil properties

Select the correct answer using the code given below.

(a) 1 and 2

(b) 4 and 5

(c) 2, 3 and 4

(d) 1, 3 and 5

2021



The black cotton soil of India has been formed due to the weathering of

- (a) Brown forest soil
- (b) Fissure volcanic rock
- (c) Granite and schist
- (d) Shale and limestone

2021



Leaf litter decomposes faster than in any other biome and as a result, the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees.” This is the most likely description of

- (a) Coniferous forest
- (b) Dry deciduous forest
- (c) Mangrove forest
- (d) Tropical rain forest

2023

Consider the following statements:

Statement-I: The soil in tropical rain forests is rich in nutrients.

Statement-II: The high temperature and moisture of tropical rainforests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the above statements?

- a) Both Statement-I and Statement-II are correct and Statement-II is Correct the explanation for Statement-I
- b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- c) Statement-I is correct but Statement-II is incorrect
- d) Statement-I is incorrect, Statement-II is correct

Questions

2020: The process of desertification does not have climatic boundaries. Justify with examples.

2020: Examine the status of forest resources of India and its resultant impact on climate change.

2019: Discuss the causes of the depletion of mangroves and explain their importance in maintaining coastal ecology.